



Answers & Solutions:

1. **D. convert the induced ac into dc**
 2. **D. electrical input power**
 3. **A. constant**
 4. **C. circuit breaker or fuse will open the circuit before too much damage is done to the motor**
 5. **D. load current in armature**
 6. **C. increase by about 10 percent**
 7. **D. series**
 8. **A. a closed one**
 9. **A. commutation**
 10. **B. pole flux**
 11. **B. approach infinity**
 12. **C. helps in energy conversion**
 13. **D. All of the above**
 14. **D. Both A and B**
 15. **B. decrease in back emf**
 16. **D. Both A and B**
 17. **C. reversed**
 18. **C. field**
 19. **C. 3**
 20. **C. neutralize cross-magnetizing flux**
 21. **D. zero**
- Solution:
- $$N_r = N_s$$
- $$\%s = \frac{N_s - N_r}{N_s} \times 100$$
- $$\%s = 0\% \rightarrow \text{Ans.}$$
22. **B. sparking**

23. A. Unfavorably

24. D. 4

Solution:

$$\text{Pole pitch} = \frac{\text{No of armature slots}}{\text{No. of poles}}$$

For two-layer winding

Number of armature slots = Number of coils

Number of armature slots = 16

$\therefore$

$$\text{Pole pitch} = \frac{16}{4}$$

Pole pitch = 4 $\rightarrow$ Ans.

25. B. open-circuit

26. B. Swinburne's

27. B. ensure the addition of emfs of consecutive turns

28. D. rotational

29. A. wave

30. B. 0.04 Wb

Solution:

The emf induced in a dc machine is given as

$$E = \frac{EP\phi}{60} \left(\frac{Z}{A} \right)$$

where : A = P (for lap wound generator)

$$E = \frac{EP\phi}{60} (Z)$$

$$200 = \frac{1200 \times \phi \times 250}{60}$$

$$\phi = 0.04 \text{ Wb} \rightarrow \text{Ans.}$$

31. C. 0.5

32. A. clockwise

33. C. the speed becomes dangerously high

34. A. armature current multiplied by back emf

35. B. load

36. B. 20 kW

Solution:

Rotor input = Air Gap Power (P_g)

$$\text{Rotor input} = \frac{\text{Rotor Cu loss}}{s}$$

$$\text{Rotor input} = \frac{600 \text{ W}}{0.03}$$

Rotor input = 20 kW $\rightarrow$ Ans.

37. A. commutator bars per pole

38. B. E_o and I_f

39. A. field flux

40. C. twice

41. C. zero

42. D. armature current

43. D. Both A and B

44. C. Total field circuit resistance must be less than the critical value

45. C. sparking

46. A. sinusoidal

47. D. to a dc source

48. A. high, low

49. A. Induced emf

50. C. full-pitch

END